

Random Walks on Graphs

Def. 1 (Markov Chain)

A stochastic process $Z = (Z_n)_{n \geq 0}$ on a countable state space V is a Markov chain if for all $n \geq 0$ and $u_0, \dots, u_{n+1} \in V$,

$$\begin{aligned}\mathbb{P}(Z_{n+1} = u_{n+1} \mid Z_n = u_n, \dots, Z_0 = u_0) \\ = \mathbb{P}(Z_{n+1} = u_{n+1} \mid Z_n = u_n).\end{aligned}$$

Def. 2 (Transition Matrix)

Let $Z = (Z_n)_{n \geq 0}$ be a Markov chain on V . Its transition matrix is $\bar{P} = (p_{u,v})_{u,v \in V}$, where

$$p_{u,v} = \mathbb{P}(Z_{n+1} = v \mid Z_n = u),$$

with $p_{u,v} \geq 0$ and $\sum_{v \in V} p_{u,v} = 1$ for all $u \in V$.

Def. 3 (Symmetric Random Walk (SRW))

Let $G = (V, E)$ be a finite or countably infinite graph with finite degrees. A symmetric random walk $(S_n)_{n \geq 0}$ on G satisfies

$$\mathbb{P}(S_{n+1} = v \mid S_n = u) = \frac{1}{\deg(u)}, \quad v \sim u.$$

Def. 4 (Reversible Markov Chain)

A Markov chain with transition matrix $P = (p_{u,v})$ is reversible with respect to $\pi : V \rightarrow (0, \infty)$ if

$$\pi_u p_{u,v} = \pi_v p_{v,u}, \quad u, v \in V.$$

Def. 5 (Weighted Graph from (P, π))

Define edge-weights

$$w_{u,v} = \pi_u p_{u,v} = \pi_v p_{v,u}, \quad W_u = \sum_{v \in V} w_{u,v}.$$

Then

$$p_{u,v} = \frac{w_{u,v}}{W_u}.$$

The pair (G, w) is called the weighted graph associated with (P, π) .

Def. 6 (Harmonic Function)

Let $U \subseteq V$. A function $f : V \rightarrow \mathbb{R}$ is harmonic on U with respect to the transition matrix P if

$$f(u) = \sum_{v \in V} p_{u,v} f(v) = \mathbb{E}(f(Z_1) \mid Z_0 = u), \quad u \in U.$$

Def. 7 (Hitting Probability)

Denote $P_u(\cdot) = \mathbb{P}(\cdot \mid Z_0 = u)$. Let $U \subseteq V$, $W = V \setminus U$, and $s \in U$. Define

$$g(u) = \mathbb{P}_u(Z_n = s \text{ for some } n < T_W),$$

where

$$T_W = \inf\{n \geq 0 : Z_n \in W\}$$

is the first-passage time to W .

Thm. 8 (Harmonicity of Hitting Probabilities)

The function g is harmonic on $U \setminus \{s\}$, with boundary conditions

$$g(s) = 1, \quad g(v) = 0 \text{ for } v \in W.$$

Def. 9 (Recurrent Markov Chain)

Let $Z = (Z_n)_{n \geq 0}$ be a Markov chain on a countable state space with distinguished state 0. Z is called *re-current* if

$$\mathbb{P}_0(\exists n \geq 1 : Z_n = 0) = 1.$$

Def. 10 (Transient Markov Chain)

Let $Z = (Z_n)_{n \geq 0}$ be a Markov chain on a countable state space with distinguished state 0. Z is called *trans-sient* if

$$\mathbb{P}_0(\exists n \geq 1 : Z_n = 0) < 1.$$

Electrical Networks

Def. 11 (Electrical Network)

Let $G = (V, E)$ be a finite connected graph with edge-weights $w : E \rightarrow (0, \infty)$. Each edge $e = \langle u, v \rangle$ has conductance w_e (resistance $1/w_e$). Let $s, t \in V$ be distinct vertices (source-set $S = \{s, t\}$).

Def. 12 (Potential Difference and Current)

For each edge $\langle u, v \rangle$, define antisymmetric quantities:

$$\phi_{u,v} = -\phi_{v,u}, \quad i_{u,v} = -i_{v,u},$$

called the potential difference from u to v , and the current from u to v , respectively. $i_{u,v}$ has the same sign as $\phi_{u,v}$.

Thm. 13 (Kirchhoff’s Potential Law)

For any cycle $v_1, \dots, v_n, v_{n+1} = v_1$,

$$\sum_{j=1}^n \phi_{v_j, v_{j+1}} = 0.$$

Thm. 14 (Kirchhoff’s Current Law)

For any $u \in V \setminus \{s, t\}$,

$$\sum_{v \in V} i_{u,v} = 0.$$

Thm. 15 (Ohm’s Law)

For any edge $\langle u, v \rangle$,

$$i_{u,v} = w_{u,v} \phi_{u,v}.$$

Def. 16 (Potential Function)

There exists $\phi : V \rightarrow \mathbb{R}$ such that for all $\langle u, v \rangle \in E$,

$$\phi_{u,v} = \phi(v) - \phi(u).$$

Thm. 17 (Harmonicity of Potentials)

Let $U = V \setminus \{s, t\}$. The potential function satisfies

$$\phi(u) = \sum_{v \in V} \frac{w_{u,v}}{W_u} \phi(v), \quad W_u = \sum_{v \in V} w_{u,v}, \quad u \in U.$$

Thm. 18 (Maximum Principle on Networks)

Let $G = (V, E)$ be a finite or infinite network with finite vertex degrees and conductances $(w_e)_{e \in E}$. Let $H = (W, F)$ be a connected subgraph and define the external boundary

$$\partial W = \{v \in V \setminus W : v \sim w \text{ for some } w \in W\}.$$

Let $\varphi : V \rightarrow [0, \infty)$ be harmonic on W , and suppose that the supremum of φ on W is attained and satisfies

$$\sup_{w \in W} \varphi(w) = \|\varphi\|_\infty := \sup_{v \in V} \varphi(v).$$

Then φ is constant on $W \cup \partial W$, and

$$\varphi(v) = \|\varphi\|_\infty \quad \forall v \in W \cup \partial W.$$

Thm. 19 (Superposition Principle)

If $i^{(1)}$ and $i^{(2)}$ satisfy Kirchhoff’s laws with the same sources, then $i^{(1)} + i^{(2)}$ also satisfies them.

Def. 20 (s/t -Flow)

An s/t -flow is a function $j = (j_{u,v})$ such that:

- $j_{u,v} = -j_{v,u}$,
- $j_{u,v} = 0$ if $u \not\sim v$,
- $\sum_{v \in V} j_{u,v} = 0$ for all $u \neq s, t$.

The flow size is $|j| = |J_s|$, where $J_u = \sum_{v \in V} j_{u,v}$.

Thm. 21 (Uniqueness of Electrical Flow)

For fixed s, t and fixed flow size, there is at most one solution to the Kirchhoff laws.

Def. 22 ($\Pi(s, a, b, t)$ -Property)

For an edge $\langle a, b \rangle$, let $\Pi(s, a, b, t)$ be the property of spanning trees that: the unique s/t path in a spanning tree passes through $\langle a, b \rangle$ in the direction $a \rightarrow b$.

Thm. 23 (Kirchhoff’s Spanning Tree Formula)

Let N be the number of spanning trees of G . For an edge $\langle a, b \rangle$, define

$$i_{a,b} = \frac{1}{N} \left(N(s, a, b, t) - N(s, b, a, t) \right),$$

where $N(s, a, b, t)$ is the number of spanning trees with property $\Pi(s, a, b, t)$. Then $(i_{a,b})$ is the unique unit s/t -flow satisfying Kirchhoff’s laws.

Thm. 24 (Uniform Spanning Tree Representation)

Let T be a spanning tree of G chosen uniformly at random. The unique unit s/t -flow satisfies, for any edge $\langle a, b \rangle$,

$$i_{a,b} = \mathbb{P}(T \text{ has } \Pi(s, a, b, t)) - \mathbb{P}(T \text{ has } \Pi(s, b, a, t)).$$

Def. 25 (Weight of a Spanning Tree)

For a weighted graph (G, w) and a spanning tree T of G , its weight is

$$w(T) = \prod_{e \in T} w_e.$$

Thm. 26 (Weighted Spanning Tree Formula)

For conductances w_e , define

$$N^* = \sum_T w(T),$$

$$N^*(s, a, b, t) = \sum_{T \text{ has } \Pi(s, a, b, t)} w(T).$$

Then

$$i_{a,b} = \frac{1}{N^*} \left(N^*(s, a, b, t) - N^*(s, b, a, t) \right)$$

is the unique unit s/t -flow.

Flows and Energy

Def. 27 (Energy of a Flow)

Let j be an s/t -flow on a weighted graph (G, w) . The (dissipated) energy of j is

$$\mathcal{E}(j) = \sum_{\langle u,v \rangle \in E} \frac{j_{u,v}^2}{w_{u,v}} = \frac{1}{2} \sum_{u,v \in V, u \sim v} \frac{j_{u,v}^2}{w_{u,v}}.$$

Prop. 28

Let $\psi : V \rightarrow \mathbb{R}$ and let j be an s/t -flow. Then

$$[\psi(t) - \psi(s)]J_s = \frac{1}{2} \sum_{u,v \in V} [\psi(v) - \psi(u)]j_{u,v}.$$

Def. 29 (Effective Conductance and Resistance)

Let i be the current satisfying the Kirchhoff laws with potential ϕ . The effective conductance W_{eff} between s and t is defined by

$$I_s = W_{\text{eff}} [\phi(t) - \phi(s)].$$

The effective resistance is

$$R_{\text{eff}} = \frac{1}{W_{\text{eff}}}.$$

Lem. 30

The effective resistance between s and t equals the en-ergy dissipated by the unit s/t -flow that satisfies the Kirchoff laws:

$$R_{\text{eff}} = \mathcal{E}(i).$$

Thm. 31 (Series Law)

Two resistors r_1, r_2 in series are equivalent to a single resistor of size

$$r_1 + r_2.$$

Thm. 32 (Parallel Law)

Two resistors r_1, r_2 in parallel are equivalent to a single resistor of size R with

$$R^{-1} = r_1^{-1} + r_2^{-1}.$$

Thm. 33 (Thomson Principle)

Among all unit s/t -flows, the unique flow satisfying the Kirchhoff laws minimizes the energy:

$$\mathcal{E}(i) = \inf\{\mathcal{E}(j) : j \text{ is a unit } s/t\text{-flow}\}.$$

Thm. 34 (Rayleigh Principle)

The effective resistance R_{eff} is a non-decreasing function of the edge resistances $(r_e)_{e \in E}$.

Recurrence and Resistance

Def. 35 (Graph Distance and Balls)

Let $0 \in V$ be a fixed origin. The graph distance is $\delta(u, v)$ = length of a shortest path from u to v . Define

$$\mathcal{B}_n = \{v \in V : \delta(0, v) \leq n\},$$

$$\partial \mathcal{B}_n = \{v \in V : \delta(0, v) = n\}.$$

Def. 36 (Truncated Network and Effective Resistance)

Let G_n be the subgraph induced by $\mathcal{B}_n$, and let $\tilde{G}_n$ be the graph obtained by identifying all vertices in $\partial \mathcal{B}_n$ into a single vertex I_n . Let $R_{\text{eff}}(n)$ be the effective resistance between 0 and I_n in $\tilde{G}_n$, and define

$$R_{\text{eff}} = \lim_{n \rightarrow \infty} R_{\text{eff}}(n).$$

Thm. 37 (Return Probability Formula)

Let Z be the reversible random walk on (G, w) starting from 0. Then

$$\mathbb{P}_0(Z_n = 0 \text{ for some } n \geq 1) = 1 - \frac{1}{W_0 R_{\text{eff}}},$$

$$W_0 = \sum_{v \sim 0} w_{0,v}.$$

Def. 38 (0/∞-Flow)

A 0/∞-flow is a vector $j = (j_{u,v})$ such that:

- $j_{u,v} = -j_{v,u}$,
- $j_{u,v} = 0$ if $u \not\sim v$,
- $\sum_{v \in V} j_{u,v} = 0$ for all $u \neq 0$.

Cor. 39 (Recurrence–Resistance Criterion)

The chain Z is recurrent if and only if $R_{\text{eff}} = \infty$.

Cor. 40 (Transience–Resistance Criterion)

The chain Z is transient if and only if there exists a non-zero 0/∞-flow j with finite energy

$$\mathcal{E}(j) = \sum_{e \in E} \frac{j_e^2}{w_e} < \infty.$$

Polya’s Theorem

Thm. 41 (Pólya’s Theorem)

Let $L^d = \mathbb{Z}^d$ be the d -dimensional cubic lattice. The symmetric random walk on L^d is

$$\begin{array}{ll} \text{recurrent if } d = 1, 2, & \text{transient if } d \geq 3. \end{array}$$

Bernoulli Bond Percolation

Def. 42 (Bernoulli Bond Percolation)

Let $G = (V, E)$ be a graph. Bernoulli bond percolation with parameter $p \in [0, 1]$ is the product measure $\mathbb{P}_p$ on $\Omega = \{0, 1\}^E$ such that each edge $e \in E$ is open with probability p and closed with probability $1 - p$, independently. The σ -algebra is generated by events depending on finitely many edges.

Def. 43 (Partial Order and Increasing Event)

For $\omega, \omega' \in \{0, 1\}^E$, define $\omega \leq \omega'$ iff $\omega(e) \leq \omega'(e)$ for all $e \in E$. A function $f : \{0, 1\}^E \rightarrow \mathbb{R}$ is *increasing* if $\omega \leq \omega' \Rightarrow f(\omega) \leq f(\omega')$. An event A is *increasing* if $\mathbf{1}_A$ is increasing. Equivalently, $\omega \in A$ and $\omega' \geq \omega$ imply $\omega' \in A$.

Lem. 44 (Monotonicity)

If $p \leq p'$, then for every increasing event A ,

$$\mathbb{P}_p(A) \leq \mathbb{P}_{p'}(A).$$

Lem. 45 (FKG Inequality)

Fix $p \in [0, 1]$. For any bounded increasing (decreasing) functions $f, g : \{0, 1\}^E \rightarrow \mathbb{R}$,

$$\mathbb{E}_p[fg] \geq \mathbb{E}_p[f]\mathbb{E}_p[g].$$

In particular, for increasing (decreasing) events A, B ,

$$\mathbb{P}_p(A \cap B) \geq \mathbb{P}_p(A)\mathbb{P}_p(B).$$

Def. 46 (Pivotal Edge)

Let A be an increasing event depending on finitely many edges $E_0 \subset E$. For $\omega \in \{0, 1\}^E$ and $e \in E_0$, let ω^e be ω with e forced open, and ω_e be ω with e forced closed. An edge e is *pivotal* for A if

$$\omega^e \in A \quad \text{but} \quad \omega_e \notin A.$$

Lem. 47 (Russo's Formula)

Let A be an increasing event depending on finitely many edges E_0 . Then

$$\frac{d}{dp} \mathbb{P}_p(A) = \sum_{e \in E_0} \mathbb{P}_p(e \text{ is pivotal for } A).$$

Def. 48 (Square Lattice and Boxes)

Identify $V(\mathbb{Z}^2) = \mathbb{Z}^2$ and

$$E(\mathbb{Z}^2) = \{\{x, y\} : |x_1 - y_1| + |x_2 - y_2| = 1\}.$$

Define the box $\Lambda_n = [-n, n]^2$. For a finite subgraph $G \subset \mathbb{Z}^2$, its boundary is

$$\partial G = \{x \in V(G) : \exists y \notin V(G) \text{ with } \{x, y\} \in E(\mathbb{Z}^2)\}.$$

Def. 49 (Percolation Probability)

For bond percolation on $\mathbb{Z}^2$, define

$$\theta(p) = \mathbb{P}_p(0 \leftrightarrow \infty),$$

the probability that 0 lies in an infinite open cluster.

Thm. 50 (Phase Transition)

For Bernoulli bond percolation on $\mathbb{Z}^2$, there exists a critical value $p_c \in (0, 1)$ such that

$$\theta(p) = 0 \quad \text{if } p < p_c, \quad \theta(p) > 0 \quad \text{if } p > p_c.$$

The regimes are called:

- subcritical if $p < p_c$,
- critical if $p = p_c$,
- supercritical if $p > p_c$.

Def. 51 (Critical Probability)

Define

$$p_c = \sup\{p : \theta(p) = 0\}.$$

Then $\theta(p) = 0$ for $p < p_c$ and $\theta(p) > 0$ for $p > p_c$.

Def. 52 (Dual Configuration)

Let $G = (V, E)$ be a plane graph with dual graph $G^* = (V^*, E^*)$. For $\omega \in \{0, 1\}^E$, define the dual configuration $\omega^* \in \{0, 1\}^{E^*}$ by

$$\omega^*(e^*) = 1 - \omega(e).$$

If $\omega \sim \mathbb{P}_p$, then $\omega^* \sim \mathbb{P}_{1-p}$ on G^* .

Thm. 53 (Infinite Cluster Dichotomy)

For Bernoulli bond percolation on $\mathbb{Z}^2$, either

- there is no infinite cluster almost surely, or
- there exists a unique infinite cluster almost surely.

Def. 54 (Translation Shift)

For $x \in \mathbb{Z}^2$, define the shift $\tau_x : \{0, 1\}^{E(\mathbb{Z}^2)} \rightarrow \{0, 1\}^{E(\mathbb{Z}^2)}$ by

$$(\tau_x \omega)(\{a, b\}) = \omega(\{a + x, b + x\}).$$

An event A is translation invariant if $\tau_x A = A$ for all $x \in \mathbb{Z}^2$.

Lem. 55 (Ergodicity of Bernoulli Percolation)

Bernoulli bond percolation on $\mathbb{Z}^2$ is ergodic. That is, if A is translation invariant, then

$$\mathbb{P}_p(A) \in \{0, 1\}.$$

Def. 56 (Disjoint Infinite Cluster Events)

For $k \geq 1$, let A_k be the event that there exist k disjoint infinite clusters. Let A_∞ be the event that infinitely many disjoint infinite clusters exist.

Thm. 57 (Uniqueness of Infinite Cluster)

For Bernoulli bond percolation on $\mathbb{Z}^2$ and any p ,

$$\mathbb{P}_p(A_k) = 0 \quad \text{for } k \geq 2, \quad \mathbb{P}_p(A_\infty) = 0.$$

Hence at most one infinite cluster exists almost surely.

Def. 58 (Edge Boundary)

For a finite set $S \subset \mathbb{Z}^2$, define the edge-boundary

$$\Delta S = \{\{x, y\} \in E(\mathbb{Z}^2) : x \in S, y \notin S\}.$$

Def. 59 (Boundary Functional)

For $p \in [0, 1]$ and a finite set $0 \in S \subset \mathbb{Z}^2$, define

$$\varphi_p(S) := p \sum_{\{x, y\} \in \Delta S} \mathbb{P}_p[0 \overset{S}{\longleftrightarrow} x].$$

Def. 60 (Critical Parameter $\tilde{p}_c$)

$$\tilde{p}_c = \sup\{p \in [0, 1] : \exists S \subset \mathbb{Z}^2, 0 \in S, \varphi_p(S) < 1\}.$$

Thm. 61 (Subcritical Exponential Decay)

Consider Bernoulli bond percolation on $\mathbb{Z}^2$.

- If $p < p_c$, then there exists $c(p) > 0$ such that for every $n \geq 1$,

$$\mathbb{P}_p[0 \leftrightarrow \partial \Lambda_n] \leq e^{-cn}.$$

- If $p > p_c$, then

$$\theta(p) = \mathbb{P}_p[0 \leftrightarrow \infty] \geq \frac{p - p_c}{p(1 - p_c)}.$$

Lem. 62 (Differential Inequality)

For $p \in [0, 1]$ and $n \geq 1$,

$$\frac{d}{dp} \mathbb{P}_p[0 \leftrightarrow \partial \Lambda_n] \geq$$

$$\frac{1}{p(1 - p)} \left(\inf_{S \subset \Lambda_n : 0 \in S} \varphi_p(S) \right) (1 - \mathbb{P}_p[0 \leftrightarrow \partial \Lambda_n]).$$

Cor. 63

For $p > \tilde{p}_c$ and every $n \geq 1$,

$$\mathbb{P}_p[0 \leftrightarrow \partial \Lambda_n] \geq \frac{p - \tilde{p}_c}{p(1 - \tilde{p}_c)}.$$

Thm. 64 (Kesten's Theorem)

For Bernoulli bond percolation on $\mathbb{Z}^2$,

$$p_c = \frac{1}{2}, \quad \theta(p_c) = 0.$$

Lem. 65

For Bernoulli bond percolation on $\mathbb{Z}^2$,

$$\theta\left(\frac{1}{2}\right) = 0.$$

Def. 66 (Rectangle Crossing Events)

For a rectangle $R = [a, b] \times [c, d]$, define

$$C_h(R) := \{\{a\} \times [c, d] \overset{R}{\longleftrightarrow} \{b\} \times [c, d]\},$$

$$C_v(R) := \{[a, b] \times \{c\} \overset{R}{\longleftrightarrow} [a, b] \times \{d\}\}.$$

Lem. 67 (Self-Dual Crossing Identity)

For $p = \frac{1}{2}$,

$$\mathbb{P}[C_v([0, n]^2)] + \mathbb{P}[C_h^*([-\frac{1}{2}, n + \frac{1}{2}] \times [\frac{1}{2}, n - \frac{1}{2}])] = 1.$$

Cor. 68

At $p = \frac{1}{2}$,

$$\mathbb{P}[C_v([0, n]^2)] \geq \frac{1}{2}.$$

Markov Chains

Def. 69 (Markov Chain)

A sequence $(X_0, X_1, \dots)$ with state space X and transition matrix P is a Markov chain if for all $x, y \in X$ and $t \geq 1$

$$\mathbb{P}(X_{t+1} = y \mid X_t = x, X_{t-1}, \dots, X_0)$$

$$= \mathbb{P}(X_{t+1} = y \mid X_t = x) = P(x, y).$$

Def. 70 (Transition Matrix)

$P = (P(x, y))_{x, y \in X}$ where

$$P(x, y) \geq 0, \quad \sum_{y \in X} P(x, y) = 1.$$

The row $P(x, \cdot)$ is the distribution of X_{t+1} given $X_t = x$.

Def. 71 (Distribution)

Let

$$\mu_t(x) = \mathbb{P}(X_t = x).$$

Then μ_t is a probability distribution on X .

Prop. 72 (Distribution Evolution)

If μ_t is the distribution of X_t , then

$$\mu_{t+1} = \mu_t P.$$

Hence

$$\mu_t = \mu_0 P^t.$$

Def. 73 (t-step Transition Probabilities)

$$P^t(x, y) = \mathbb{P}_x(X_t = y).$$

Def. 74 (Point Mass Initial Distribution)

For $x \in X$ define

$$\delta_x(y) = \begin{cases} 1 & y = x \\ 0 & y \neq x. \end{cases}$$

Then

$$\mathbb{P}_x(X_t = y) = (\delta_x P^t)(y) = P^t(x, y).$$

Def. 75 (Stationary Distribution)

A distribution π is stationary if

$$\pi = \pi P.$$

Def. 76 (Expectation Operator)

For a function $f : X \rightarrow \mathbb{R}$,

$$Pf(x) = \sum_{y \in X} P(x, y)f(y) = \mathbb{E}_x[f(X_1)].$$

Rem. 77 (Two-State Frog Chain)

Let $X = \{e, w\}$ and

$$P = \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix}.$$

If $\mu_t = (\mu_t(e), \mu_t(w))$ then

$$\mu_{t+1} = \mu_t P.$$

The stationary distribution satisfies

$$\pi = \pi P,$$

giving

$$\pi(e) = \frac{q}{p+q}, \quad \pi(w) = \frac{p}{p+q}.$$

Define

$$\Delta_t = \mu_t(e) - \frac{q}{p+q}.$$

Then

$$\Delta_{t+1} = (1 - p - q)\Delta_t$$

and hence

$$\Delta_t = (1 - p - q)^t \Delta_0.$$

Thus if $0 < p < 1$ and $0 < q < 1$,

$$\lim_{t \rightarrow \infty} \mu_t(e) = \frac{q}{p+q}, \quad \lim_{t \rightarrow \infty} \mu_t(w) = \frac{p}{p+q}.$$

The convergence rate is determined by the eigenvalue $1 - p - q$ of P .

Def. 78 (Random Walk on the n -Cycle)

Let $X = \mathbb{Z}_n = \{0, 1, \dots, n-1\}$. The random walk on the n -cycle has transition matrix

$$P(j, k) = \begin{cases} \frac{1}{2} & k \equiv j+1 \pmod{n}, \\ \frac{1}{2} & k \equiv j-1 \pmod{n}, \\ 0 & \text{otherwise.} \end{cases}$$

Equivalently, let $Z \in \{-1, 1\}$ with $\mathbb{P}(Z = 1) = \mathbb{P}(Z = -1) = \frac{1}{2}$. Then

$$X_{t+1} = X_t + Z \pmod{n}.$$

Def. 79 (Random Mapping Representation)

A random mapping representation of a transition matrix P on state space X is a function

$$f : X \times \Lambda \rightarrow X$$

together with a Λ -valued random variable Z such that

$$\mathbb{P}\{f(x, Z) = y\} = P(x, y).$$

If $Z_1, Z_2, \dots$ are i.i.d. with the same distribution as Z and

$$X_n = f(X_{n-1}, Z_n), \quad n \geq 1,$$

then $(X_0, X_1, \dots)$ is a Markov chain with transition matrix P .

Prop. 80

Every transition matrix on a finite state space admits a random mapping representation.

Rem. 81

Random mapping representations are not unique. They are useful for simulation of Markov chains and for coupling multiple trajectories using the same auxiliary randomness.

Def. 82 (Irreducible Markov Chain)

A Markov chain with transition matrix P on state space X is irreducible if for all $x, y \in X$ there exists $t \geq 1$ such that

$$P^t(x, y) > 0.$$

That is, every state can be reached from every other state with positive probability.

Def. 83 (Return Times)
For $x \in X$, define

$$\mathcal{T}(x) = \{t \geq 1 : P^t(x, x) > 0\}.$$

This is the set of times at which the chain can return to x .

Def. 84 (Period)
The period of a state x is

$$\gcd \mathcal{T}(x).$$

Lem. 85
If P is irreducible, then

$$\gcd \mathcal{T}(x) = \gcd \mathcal{T}(y)$$

for all $x, y \in X$.

Def. 86 (Aperiodic Chain)
An irreducible Markov chain is aperiodic if all states have period 1. If the period is greater than 1, the chain is periodic.

Lem. 87 (Schur Lemma)
Let $S \subset \mathbb{Z}_{\geq 1}$ and let $g = \gcd(S)$. Then $\exists m_0$ such that for all $m \geq m_0$,

$$mg = \sum_{s \in S} a_s s \quad \text{with } a_s \in \mathbb{Z}_{\geq 0}.$$

Prop. 88
If P is irreducible and aperiodic, then there exists r_0 such that

$$P^r(x, y) > 0$$

for all $x, y \in X$ and all $r \geq r_0$.

Def. 89 (Lazy Chain)
Given a transition matrix P , define

$$Q = \frac{I + P}{2}.$$

Then Q is called the lazy version of P .

For a lazy chain, $Q(x, x) > 0$ for all $x \in X$, so the chain is aperiodic.

Lazy Random Walk on $\mathbb{Z}_n$: For $n \geq 3$, the lazy random walk transition matrix is

$$Q(j, k) = \begin{cases} \frac{1}{4} & k \equiv j + 1 \pmod{n}, \\ \frac{1}{2} & k \equiv j \pmod{n}, \\ \frac{1}{4} & k \equiv j - 1 \pmod{n}, \\ 0 & \text{otherwise.} \end{cases}$$

This chain is irreducible and aperiodic.

Def. 90 (Graph)
A graph $G = (V, E)$ consists of a vertex set V and an edge set

$$E \subset \{\{x, y\} : x, y \in V, \ x \neq y\}.$$

If $\{x, y\} \in E$, we write $x \sim y$ and say that x and y are neighbors.

Def. 91 (Degree)
The degree of a vertex x is

$$\deg(x) = |\{y \in V : y \sim x\}|.$$

Def. 92 (Simple Random Walk on a Graph)
Given a graph $G = (V, E)$, the simple random walk on G is the Markov chain with state space V and transition matrix

$$P(x, y) = \begin{cases} \frac{1}{\deg(x)} & y \sim x, \\ 0 & \text{otherwise.} \end{cases}$$

When the chain is at vertex x , it chooses one of the neighbors of x uniformly at random and moves to that vertex.

Def. 93 (Stationary Distribution)
A probability distribution π on the state space X is stationary if

$$\pi = \pi P.$$

Equivalently,

$$\pi(y) = \sum_{x \in X} \pi(x) P(x, y) \quad \forall y \in X.$$

If the chain starts with $\mu_0 = \pi$, then $\mu_t = \pi$ for all $t \geq 0$.

Rem. 94 (Random Walk on a Graph)
For simple random walk on a graph $G = (V, E)$,

$$\pi(y) = \frac{\deg(y)}{2|E|}$$

is a stationary distribution.

If the graph is d -regular, then

$$\pi(y) = \frac{1}{|V|}$$

(the uniform distribution) is stationary.

Def. 95 (Hitting Time)
For $x \in X$, the hitting time of x is

$$\tau_x = \min\{t \geq 0 : X_t = x\}.$$

Def. 96 (First Return Time)

$$\tau_x^+ = \min\{t \geq 1 : X_t = x\}.$$

If $X_0 = x$, this is the first return time to x .

Lem. 97
For any states x, y in a finite irreducible Markov chain,

$$E_x(\tau_y^+) < \infty.$$

Prop. 98 (Stationary Distribution Construction)
Fix $z \in X$ and define

$$\begin{aligned} \tilde{\pi}(y) &= E_z(\text{number of visits to } y \text{ before returning to } z) \\ &= \sum_{t=0}^{\infty} P_z(X_t = y, \tau_z^+ > t). \end{aligned}$$

If $E_z(\tau_z^+) < \infty$, then

$$\pi(y) = \frac{\tilde{\pi}(y)}{E_z(\tau_z^+)}$$

is a stationary distribution.

Def. 99 (Harmonic Function)
A function $h : X \rightarrow \mathbb{R}$ is harmonic at x if

$$h(x) = \sum_{y \in X} P(x, y) h(y).$$

Equivalently,

$$Ph = h.$$

Lem. 100
If P is irreducible and h is harmonic on X , then h is constant.

Cor. 101 (Uniqueness of Stationary Distribution)
If P is irreducible and X is finite, then there exists a unique probability distribution π such that

$$\pi = \pi P.$$

Prop. 102 (Mean Return Time Formula)
For an irreducible Markov chain with stationary distribution π ,

$$\pi(z) = \frac{1}{E_z(\tau_z^+)}.$$

Def. 103 (Total Variation Distance)
For probability distributions μ, ν on X ,

$$\|\mu - \nu\|_{\text{TV}} = \max_{A \subset X} |\mu(A) - \nu(A)|.$$

Prop. 104

$$\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_{x \in X} |\mu(x) - \nu(x)|.$$

Rem. 105
Let $B = \{x : \mu(x) \geq \nu(x)\}$. Then

$$\|\mu - \nu\|_{\text{TV}} = \sum_{x \in B} (\mu(x) - \nu(x)).$$

Prop. 106 (Dual Characterisation)

$$\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sup_{\|f\|_{\infty} \leq 1} \left| \sum_x f(x) \mu(x) - \sum_x f(x) \nu(x) \right|.$$

Rem. 107

$$\|\mu - \nu\|_{\text{TV}} \leq \|\mu - \eta\|_{\text{TV}} + \|\eta - \nu\|_{\text{TV}}.$$

Rem. 108
If $X = \{e, w\}$, then

$$\|\mu_t - \pi\|_{\text{TV}} = |\mu_t(e) - \pi(e)|.$$

Def. 109 (Coupling)
A coupling of probability distributions μ, ν on X is a pair (X, Y) on a common probability space such that

$$\mathbb{P}(X = x) = \mu(x), \quad \mathbb{P}(Y = y) = \nu(y).$$

Equivalently, a joint distribution $q(x, y)$ on $X \times X$ with marginals μ, ν .

Rem. 110
Any μ, ν admit an independent coupling: $q(x, y) = \mu(x)\nu(y)$.

Prop. 111 (Coupling Characterisation)

$\|\mu - \nu\|_{\text{TV}} = \inf\{\mathbb{P}(X \neq Y) : (X, Y) \text{ is a coup. of } \mu, \nu\}.$

Rem. 112 (Optimal Coupling)
There exists a coupling such that

$$\mathbb{P}(X \neq Y) = \|\mu - \nu\|_{\text{TV}}.$$

Rem. 113 (Maximal Overlap)
Define

$$p = \sum_{x \in X} \min\{\mu(x), \nu(x)\}.$$

Then

$$p = 1 - \|\mu - \nu\|_{\text{TV}}.$$

$\|\mu - \nu\|_{\text{TV}}$ is the smallest probability that two random variables with laws μ, ν can differ.

Thm. 114 (Convergence Theorem)
Let P be an irreducible and aperiodic transition matrix on a finite state space, with stationary distribution π . Then $\exists \alpha \in (0, 1)$ and $C > 0$ such that

$$\max_{x \in X} \|P^t(x, \cdot) - \pi\|_{\text{TV}} \leq C\alpha^t.$$

Rem. 115
Hence for any initial distribution μ ,

$$\|\mu P^t - \pi\|_{\text{TV}} \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

Def. 116

$$d(t) := \max_{x \in X} \|P^t(x, \cdot) - \pi\|_{\text{TV}},$$

$$\bar{d}(t) := \max_{x, y \in X} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}}.$$

Lem. 117

$$d(t) \leq \bar{d}(t) \leq 2d(t).$$

Def. 118
Let μ be a probability distribution on X . Define

$$\mu P^t := \sum_{x \in X} \mu(x) P^t(x, \cdot).$$

Prop. 119

$$d(t) = \sup_{\mu} \|\mu P^t - \pi\|_{\text{TV}}, \quad \bar{d}(t) = \sup_{\mu, \nu} \|\mu P^t - \nu P^t\|_{\text{TV}}.$$

Lem. 120 (Submultiplicativity)

$$\bar{d}(s + t) \leq \bar{d}(s) \bar{d}(t).$$

Rem. 121
 $\bar{d}(t)$ is non-increasing in t .

Rem. 122
For positive integers c, t ,

$$d(ct) \leq \bar{d}(ct) \leq \bar{d}(t)^c.$$

Rem. 123
 $d(t)$ measures distance to stationarity, while $\bar{d}(t)$ measures worst-case distance between two starting states.

Def. 124 (Mixing Time)

$$t_{\text{mix}}(\varepsilon) := \min\{t : d(t) \leq \varepsilon\}, \quad t_{\text{mix}} := t_{\text{mix}}(1/4).$$

Rem. 125
For $\ell \in \mathbb{N}$,

$$d(\ell t_{\text{mix}}(\varepsilon)) \leq (2\varepsilon)^\ell.$$

In particular,

$$d(\ell t_{\text{mix}}) \leq 2^{-\ell}.$$

Rem. 126

$$t_{\text{mix}}(\varepsilon) \leq \lceil \log_2(\varepsilon^{-1}) \rceil t_{\text{mix}}.$$

Rem. 127
 $t_{\text{mix}}(\varepsilon)$ is the time needed for the chain to be within ε of stationarity in total variation distance.

Def. 128 (Coupling of Markov Chains)
A coupling of a Markov chain with transition matrix P is a process $(X_t, Y_t)_{t \geq 0}$ such that both (X_t) and (Y_t) are Markov chains with transition matrix P .

Def. 129 (Markovian Coupling)
A Markovian coupling is a Markov chain (X_t, Y_t) on $X \times X$ such that for all $x, y \in X$,

$$\mathbb{P}(X_{t+1} = x' \mid X_t = x, Y_t = y) = P(x, x'),$$

$$\mathbb{P}(Y_{t+1} = y' \mid X_t = x, Y_t = y) = P(y, y').$$

Def. 130
If (X_t) and (Y_t) are coupled Markov chains with $X_0 = x$ and $Y_0 = y$, we write $\mathbb{P}_{x, y}$ for probabilities on the joint space. $\mathbb{P}_{x, y}$ denotes probabilities under the coupling started from (x, y) .

Thm. 131 (Coalescence Time)
Let (X_t, Y_t) be a coupling of a Markov chain with transition matrix P , with $X_0 = x, Y_0 = y$. Define the coalescence time

$$\tau_{\text{couple}} := \min\{t : X_s = Y_s \ \forall s \geq t\}.$$

Then

$$\|P^t(x, \cdot) - P^t(y, \cdot)\|_{\text{TV}} \leq \mathbb{P}_{x, y}(\tau_{\text{couple}} > t).$$

Cor. 132

$$d(t) \leq \max_{x, y} \mathbb{P}_{x, y}(\tau_{\text{couple}} > t).$$

Cor. 133 (Mixing Time Bound)

$$t_{\text{mix}} \leq 4 \max_{x, y} \mathbb{E}_{x, y}(\tau_{\text{couple}}).$$

Rem. 134 (Lazy Random Walk on $\{0, 1\}^n$)
Choose a coordinate uniformly and refresh it with an independent fair bit. Using the natural coupling, if τ is the first time all coordinates have been selected, then

$$d(n \log n + cn) \leq \mathbb{P}(\tau > n \log n + cn) \leq e^{-c}.$$

Hence

$$t_{\text{mix}}(\varepsilon) \leq n \log n + n \log(1/\varepsilon), \quad t_{\text{mix}} = O(n \log n).$$

Rem. 135 (Lazy Biased Walk on $\mathbb{Z}_n$)
For the lazy $(p - q)$ -biased walk on $\mathbb{Z}_n$,

$$\frac{n^2}{32} \leq t_{\text{mix}} \leq n^2.$$

Rem. 136 (Lazy Random walk on $\mathbb{Z}_n^d$)
For the lazy random walk on the d -dimensional torus,

$$t_{\text{mix}}(\varepsilon) \leq dn^2 \lceil \log_4(d/\varepsilon) \rceil.$$

In particular,

$$t_{\text{mix}} \leq d^2 n^2.$$

Def. 137 (Rooted b -ary Tree)
A rooted finite b -ary tree of depth k , denoted $T_{b,k}$, is a tree with root ρ such that: the root has degree b , vertices at depths $1, \dots, k - 1$ have degree $b + 1$, and vertices at depth k are leaves.

Rem. 138 (Lazy Random Walk on the Finite Binary Tree)

For the lazy random walk on $T_{2,k}$ with $n = (2^{k+1} - 1)$ vertices, a coupling argument gives

$$t_{\text{mix}} \leq 16n.$$

Rem. 139 (Winning Streak Chain)
State space $\{0, \dots, n\}$, with transitions

$$P(i, 0) = \frac{1}{2} \quad (0 \leq i \leq n),$$

$$P(i, i + 1) = \frac{1}{2} \quad (0 \leq i < n), \quad P(n, n) = \frac{1}{2}.$$

Its stationary distribution is

$$\pi(i) = \begin{cases} 2^{-(i+1)}, & 0 \leq i \leq n - 1, \\ 2^{-n}, & i = n. \end{cases}$$

Using the natural coupling,

$$d(t) \leq 2^{-t}, \quad t_{\text{mix}}(\varepsilon) \leq \lceil \log_2(1/\varepsilon) \rceil.$$

In particular,

$$t_{\text{mix}} \leq 2.$$

Prop. 140
Let Q be irreducible and let $P = (Q + I)/2$ be the lazy chain. Then

$$\|P^t(x, \cdot) - P^{t+1}(x, \cdot)\|_{\text{TV}} \leq \frac{1}{\sqrt{t}}.$$

Martingales

Def. 141 (Martingale)
Let $(\mathcal{F}_n)_{n \geq 0}$ be a filtration and $(Y_n)_{n \geq 0}$ be adapted to it. Then $(Y_n, \mathcal{F}_n)$ is a martingale if for all $n \geq 0$,

- $\mathbb{E}[|Y_n|] < \infty$,
- $\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] = Y_n$.

Def. 142 (Submartingale and Supermartingale)
Let $(\mathcal{F}_n)_{n \geq 0}$ be a filtration and $(Y_n)_{n \geq 0}$ be adapted to it.

It is a submartingale if for all $n \geq 0$,

- $\mathbb{E}[Y_n^+] < \infty$,
- $\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] \geq Y_n$.

It is a supermartingale if for all $n \geq 0$,

- $\mathbb{E}[Y_n^-] < \infty$,
- $\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] \leq Y_n$.

Rem. 143
For $X \in \mathbb{R}$,

$$X^+ = \max\{0, X\}, \quad X^- = -\min\{0, X\},$$

so that

$$X = X^+ - X^-, \quad |X| = X^+ + X^-.$$

Y is a martingale iff it is both a submartingale and a supermartingale.

Y is a submartingale iff $-Y$ is a supermartingale.

Def. 144 (Predictable Process)
A process $(S, \mathcal{F}) = \{(S_n, \mathcal{F}_n) : n \geq 0\}$ is predictable if S_n is $\mathcal{F}_{n-1}$ -measurable for all $n \geq 1$. It is increasing if

$$S_0 = 0, \quad \mathbb{P}(S_n \leq S_{n+1}) = 1 \quad \forall n.$$

Thm. 145 (Doob Decomposition)
Let $(Y, \mathcal{F})$ be a submartingale with finite means. Then there exist a martingale $(M, \mathcal{F})$ and an increasing predictable process $(A, \mathcal{F})$ such that

$$Y_n = M_n + A_n, \quad A_0 = 0,$$

and this decomposition is unique. Moreover,

$$A_n = \sum_{k=0}^{n-1} \mathbb{E}[Y_{k+1} - Y_k \mid \mathcal{F}_k],$$

Def. 146 (Compensator)
In the Doob decomposition

$$Y_n = M_n + S_n,$$

the process $(S, \mathcal{F})$ is called the compensator of $(Y, \mathcal{F})$.

Thm. 147
Let $(X_k)_{k \geq 0}$ be an irreducible, aperiodic Markov chain on a finite state space with stationary distribution π , started from stationarity. For $x \in \Omega$, define the local time

$$L_n(x) := \sum_{k=0}^{n-1} \mathbf{1}_{\{X_k=x\}}.$$

Then there exist constants $c, C > 0$ such that for all $n \geq 1$ and $\varepsilon \geq 0$,

$$\mathbb{P}\left(\left|\frac{L_n(x)}{n} - \pi(x)\right| \geq \varepsilon\right) \leq Ce^{-cn\varepsilon^2}.$$

Equivalently,

$$\mathbb{P}(|L_n(x) - n\pi(x)| \geq u) \leq C \exp\left(-c\frac{u^2}{n}\right).$$

Thm. 148 (Abracadabra Theorem)
Let $\xi_1, \xi_2, \dots$ be i.i.d. letters on an alphabet $\mathcal{A}$, with

$$\mathbb{P}(\xi_1 = a) = p_a.$$

Fix a target word

$$w = a_1 a_2 \cdots a_m,$$

and let

$$\tau_w := \min\{n \geq m : \xi_{n-m+1} \cdots \xi_n = w\}$$

be the first occurrence time of w . Then

$$\mathbb{E}[\tau_w] = \sum_{r=1}^m \frac{\mathbf{1}_{\{a_1 \cdots a_r = a_{m-r+1} \cdots a_m\}}}{p_{a_1} \cdots p_{a_r}}.$$

That is, one sums over all overlap lengths r for which the prefix of length r equals the suffix of length r .

Thm. 149 (Azuma–Hoeffding Inequality)
Let $(Y, \mathcal{F})$ be a martingale, and suppose there exist real numbers $K_1, K_2, \dots$ such that

$$\mathbb{P}(|Y_n - Y_{n-1}| \leq K_n) = 1 \quad \forall n.$$

Then for $x > 0$,

$$\mathbb{P}(|Y_n - Y_0| > x) \leq 2 \exp\left(-\frac{x^2}{2 \sum_{i=1}^n K_i^2}\right).$$

Thm. 150 (Martingale Central Limit Theorem)
Let $(D_{i,n})_{1 \leq i \leq n}$ be a triangular array such that:

- there exists a triangular array $(X_{i,n})_{1 \leq i \leq n}$ with $(X_{1,n}, \dots, X_{n,n})$ independent and $D_{i,n}$ is a function of $(X_{1,n}, \dots, X_{i,n})$;
-

$$\mathbb{E}[D_{i,n} \mid X_{1,n}, \dots, X_{i-1,n}] = 0;$$

-
-
- $$\sum_{i=1}^n \mathbb{E}[D_{i,n}^2 \mid X_{1,n}, \dots, X_{i-1,n}] = 1;$$

-
-
-
- (Lindeberg's condition) for every $\delta > 0$,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{E}\left[D_{i,n}^2 \mathbf{1}_{\{|D_{i,n}| > \delta\}}\right] = 0.$$

Then, with

$$Y_n = \sum_{i=1}^n D_{i,n},$$

we have

$$Y_n \xrightarrow{d} \mathcal{N}(0, 1).$$

Useful Results

Thm. 151 (Markov Inequality)

Let $X \geq 0$ and $a > 0$. Then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Thm. 152 (Chebyshev Inequality)

Let X have mean μ and variance σ^2 . Then for $a > 0$,

$$\mathbb{P}(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}.$$

Thm. 153 (Jensen Inequality)

Let φ be convex and X integrable. Then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

Tutorial Results

Rem. 154 (Variance Bound)

If $\text{Var}(Z_n) = n$, then for every $a > 0$,

$$\mathbb{P}(|Z_n| \geq a\sqrt{n}) \leq \frac{1}{a^2}$$

by Chebyshev. In particular, for every $\varepsilon > 0$,

$$\mathbb{P}\left(|Z_n| \leq \frac{\sqrt{n}}{\sqrt{\varepsilon}}\right) \geq 1 - \varepsilon.$$

So $Z_n = O(\sqrt{n})$ with high probability.

Rem. 155 (Discrete Laplacian on $\varepsilon\mathbb{Z}^d$)

For SSRW on $V = \varepsilon L_{\lfloor \nu/\varepsilon \rfloor}^d$,

$$\varphi(x) = \frac{1}{2d} \sum_{j=1}^d (\varphi(x + \varepsilon e_j) + \varphi(x - \varepsilon e_j))$$

iff

$$\Delta_\varepsilon \varphi(x) := \frac{1}{\varepsilon^2} \sum_{j=1}^d (\varphi(x + \varepsilon e_j) + \varphi(x - \varepsilon e_j) - 2\varphi(x)) = 0.$$

Thus harmonic functions for P are discretizations of harmonic functions.

Thm. 156 (Maximum Principle on Networks)

If φ is harmonic on U , then

$$\sup_{u \in U \cup \partial U} \varphi(u) = \sup_{u \in \partial U} \varphi(u),$$

$$\partial U = \{v \in V \setminus U : \exists u \in U, \langle uv \rangle \in E\}.$$

Equivalently, a harmonic function cannot attain a strict interior maximum unless constant.

Rem. 157 (Identifying vertices)

If G' is obtained from G by identifying v_1, v_2 , then

$$R_{\text{eff}}(G', w') = \lim_{w_e \rightarrow +\infty} R_{\text{eff}}(G, w),$$

where $e = \langle v_1 v_2 \rangle$. This corresponds to shorting the edge e .